

Problem Statement & Motivation

In today's markets, price action is increasingly driven by "narrative." However, for retail investors, this narrative is fragmented across financial news sites, social sentiment, and complex institutional data like options flow. A typical investor spends over 2 hours daily cross-referencing news headlines with price charts, often missing the "divergence", where news is positive but institutional positioning (options) suggests a downturn.

Target Users

- **Retail Swing Traders:** Individuals who need a daily "bias check" before the market opens.
- **Student Investors:** Users learning to synthesize news with technical market indicators.

Why an Agentic AI Solution (via Google Opal)?

Traditional dashboards only display data. An agentic solution built in **Google Opal** is superior because it:

- **Reasons across steps:** It doesn't just show news; it compares the "News" output with the "Market Data " to see if they agree.
- **Lowers the barrier:** Using Opal's "vibe coding" allows for rapid prototyping of complex logic, like a "Critic Agent" that flags biased reporting, without needing a deep software engineering background.
- **Orchestrates Workflows:** It can autonomously decide to perform a "Deep Research" step if the initial news summary is too vague.

Use Cases & Scenarios

Scenario 1: The Daily "Market Pulse"

- **User Goal:** A user wants a 60-second summary of the S&P 500 and Nasdaq sentiment.
- **Agentic Tasks:** The system uses a **Search Agent** (Search node) to find the top 5 macro headlines, a **Sentiment Agent** (Gemini node) to score them, and a **Summary Agent** to produce a "Market Bias" report.

Scenario 2: The "Divergence" Check for Specific Tickers

- **User Goal:** A user is bullish on a stock (e.g., AAPL) and wants to see if the options flow matches the news.
- **Agentic Tasks:** A **News Agent** retrieves recent articles while a **Data Analyst Agent** (via Yahoo finance) pulls options positioning. A **Reviewer Agent** then highlights if the news is bullish but the flow is bearish, providing a "Contrarian Warning."

Scenario 3: Automated Research Logging

- **User Goal:** A user wants to save all bullish trade ideas to a central location.
- **Agentic Tasks:** The workflow identifies high-conviction signals and uses an **Output Agent** (Google Sheets/Docs node) to automatically log the ticker, sentiment score, and justification for future review.

Initial System Design

The system will be built using **Google Opal's visual workflow editor**, chaining multiple "functional agents" represented by logic nodes.

Agentic Workflow & Roles

- **Planner Agent (Opal Orchestrator):** The entry point that takes the user's ticker or market query and routes it.

- **The Scout (Search/Tools):** Uses Google Search and Web Retrieval nodes to fetch live data and news.
- **The Financial Analyst (Gemini Node):** Interprets news text and assigns a sentiment score (extremely bullish, bullish , neutral, bearish , extremely bearish)
- **The Market Critic (Logic/Reviewer Node):** Compares results from different sources. If the Sentiment Agent and Data Agent disagree, the Critic forces a "Reasoning Trace" to explain why.
- **The Clerk (Output):** Formats the final dashboard and logs data to Google Sheets.

Tools & UI

- **Frontend:** Opal's built-in **App View** will serve as the user interface, providing a clean, form-based input and a formatted report output.
- **Integrations:** Google Search , Google Sheets (for data persistence), Yahoo Finance (for options data) and Gemini 1.5 Pro(logic engine).

Evaluation Plan

To ensure the system is useful and accurate, I will evaluate it using the following metrics:

1. **Sentiment Accuracy:** Comparing the AI's "Bullish/Bearish" rating against three professional analyst reports for the same ticker.
2. **Source Reliability:** Tracking the percentage of "hallucinated" or irrelevant news articles fetched by the Scout agent.
3. **Conflict Resolution:** Measuring how often the "Critic Agent" successfully identifies a divergence between news sentiment and market data.
4. **User Efficiency:** Measuring the time taken to generate a summary via the dashboard vs. manual searching.

Feasibility, Risks, & Timeline

Risks

- **Hallucination:** Financial data requires precision
- **Interpretation:** Financial data can be interpreted differently by different investors
- **Mitigation:** Strict prompting in Opal nodes to only use provided search snippets.
- **Opal Tool Constraints:** Opal is no-code; complex custom API calls for real-time options might be limited.
- **Mitigation:** Use Google Search to find reputable aggregate options flow reports instead of raw API streams.

Timeline

- **Week 1-2:** Setup Opal environment and define "Scout" and "Analyst" nodes.
- **Week 3-4 (Milestone II):** Implement the multi-agent logic (Reviewer/Critic nodes) and "Conflict Detection."
- **Week 5-6:** Build the front-end UI in App View and connect to Google Sheets
- **Week 7:** Conduct the Evaluation (testing against analyst reports).
- **Week 8:** Final Presentation and Documentation.